

Problem 1.1.

1. For $n \in \mathbb{N}$ define $B_n^{(1)} = A_n$ and $B_n^{(2)} = A_n^c$. Then for each $k \in \mathbb{N}$ the family of 2^k events

$$E_{i_1, \dots, i_k} := \bigcap_{n=1}^k B_n^{(i_n)}, \quad i_n \in \{1, 2\}$$

forms a disjoint partition of Ω and, by independence, each event has probability 2^{-k} . Now any $\omega \in \Omega$ is, for each $k \in \mathbb{N}$, contained in such an event and thereby

$$P(\{\omega\}) \leq 2^{-k} \text{ for all } k \in \mathbb{N} \implies P(\{\omega\}) = 0,$$

so no point masses can exist.

2. Taking the same definitions of a), defining $\beta_n = \max\{p_n, 1 - p_n\}$ and using the basic inequality $1 - x \leq e^{-x}$

$$P(E_{i_1, \dots, i_k}) \leq \prod_{n=1}^k \beta_n \leq \exp\left(-\sum_{n=1}^k (1 - b_n)\right) = \exp\left(-\sum_{n=1}^k \alpha_n\right)$$

and the right hand side converges to 0 as $k \rightarrow \infty$ so we can use the same argumentation as in a).